

Bayesian quantile correction and synthesis for climate products

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QUESTION

Correction before synthesis

MODEL

One state-space model for correction and synthesis

How can historical archives estimate source corrections before forecasts with different horizons are blended into one predictive distribution?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge, modeled on the $\log(1 + x)$ scale with x in m^3/s . Before each origin, USGS observations and retrospective ECMWF/GloFAS and NOAA/NWS products identify time-varying source corrections. After the origin, issued ensembles and exogenous information produce source-adjusted predictive flow quantiles.



USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.



ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

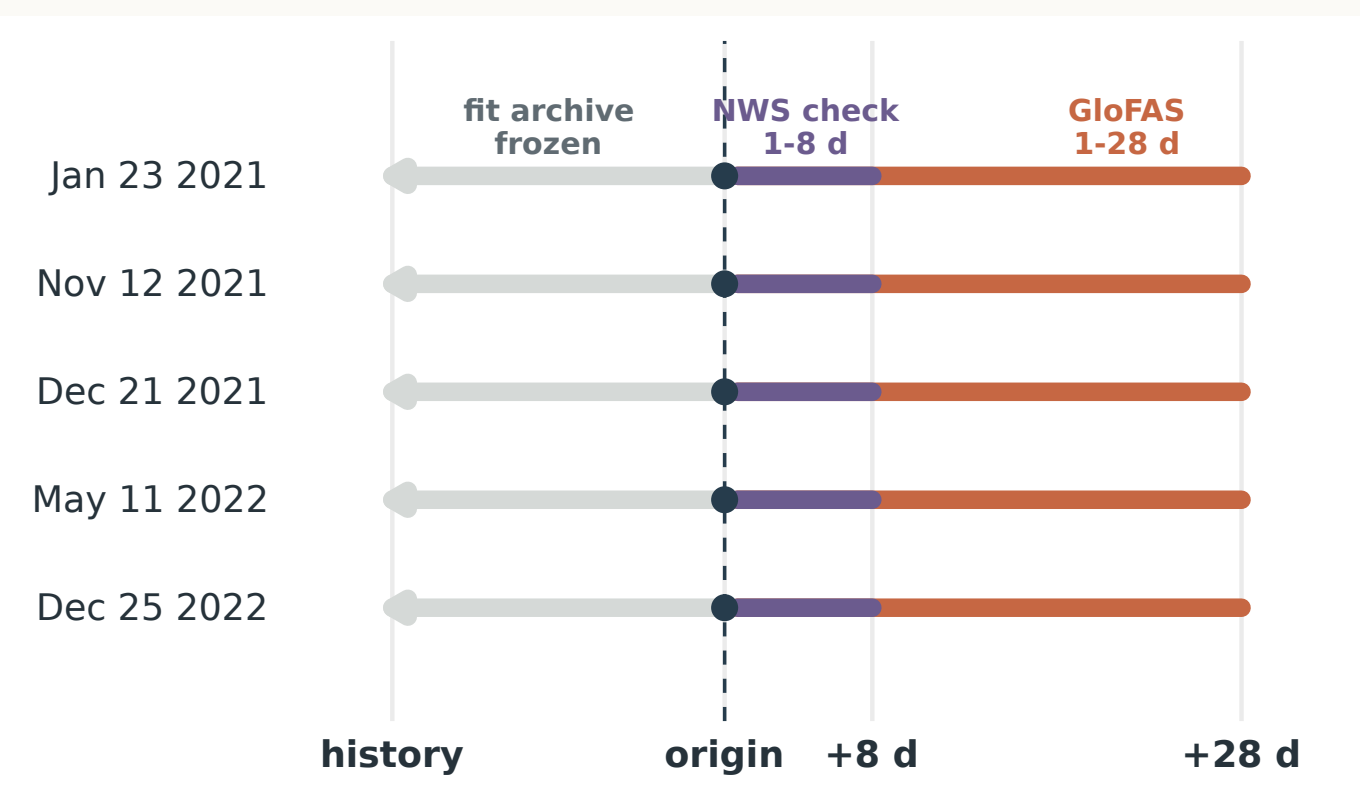


NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

VALIDATION

Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent Dynamic Quantile Linear Model (DQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7
quantile lanes

≤12,995
USGS archive days

~2–4 h
all lanes + synthesis

largest archive shown is the Dec 25 2022 origin; wall-clock on 64 cores, workflow/output-retention dependent.

Parallel quantile lanes
Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

Posterior approximation
Mean-field variational Bayes (MFVB) retains closed-form Gaussian filtering and smoothing within latent-state blocks while approximating posterior dependence across blocks.

Non-conjugate updates
Variational Bayes with Laplace–Delta (VB-LD) uses local modes, curvature, and Delta-method moments for non-conjugate extended asymmetric Laplace (exAL) scale and skewness terms.

State evolution
Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate adjustments.

Source-aware quantile correction and synthesis
Retrospective products identify source corrections before the origin; forecast products enter through those corrected channels.

Correction before synthesis. Before the origin, observations and retrospective products identify the shared quantile state and source corrections. After the origin, issued forecast products and exogenous covariates update the same state to form source-adjusted predictive quantiles.

Expanded quantile state-space model
For each target quantile p_0 and forecast origin T , observed flow, retrospective product histories, and issued forecast members are linked through one latent river-flow quantile. Retrospective products identify source corrections before the origin; forecast products enter through those corrected source channels.

Observed measurements: $y_t^o \sim \text{exAL}_{p_0}(\mathbf{F}_t^o \theta_t + \zeta_t, \sigma^o, \gamma^o), \quad 1 \leq t \leq T,$

Retrospective products: $z_t^j \sim \text{exAL}_{p_0}(\mathbf{F}_t^j(\theta_t + \delta_t^j) + \zeta_t, \sigma^j, \gamma^j), \quad j \in \mathcal{J}, \quad 1 \leq t \leq T,$

Forecast products: $y_{T+k}^j(k) \sim \text{exAL}_{p_0}(\mathbf{F}_{T+k}^j(\theta_{T+k} + \delta_{T+k}^j) + \zeta_{T+k}, \sigma^j, \gamma^j), \quad j \in \mathcal{J}, \quad i \in \mathcal{I}_j, \quad k \in \mathcal{K}_j,$

Trend and seasonal states: $\theta_t \mid \theta_{t-1} \sim \mathcal{N}(\mathbf{G}_t \theta_{t-1}, \mathbf{W}_t^{\theta}), \quad 1 \leq t \leq T + K_{\max},$

Source correction states: $\delta_t^j \mid \delta_{t-1}^j \sim \mathcal{N}(\mathbf{G}_t \delta_{t-1}^j, \mathbf{W}_t^{\delta}), \quad j \in \mathcal{J}, \quad 1 \leq t \leq T + K_{\max},$

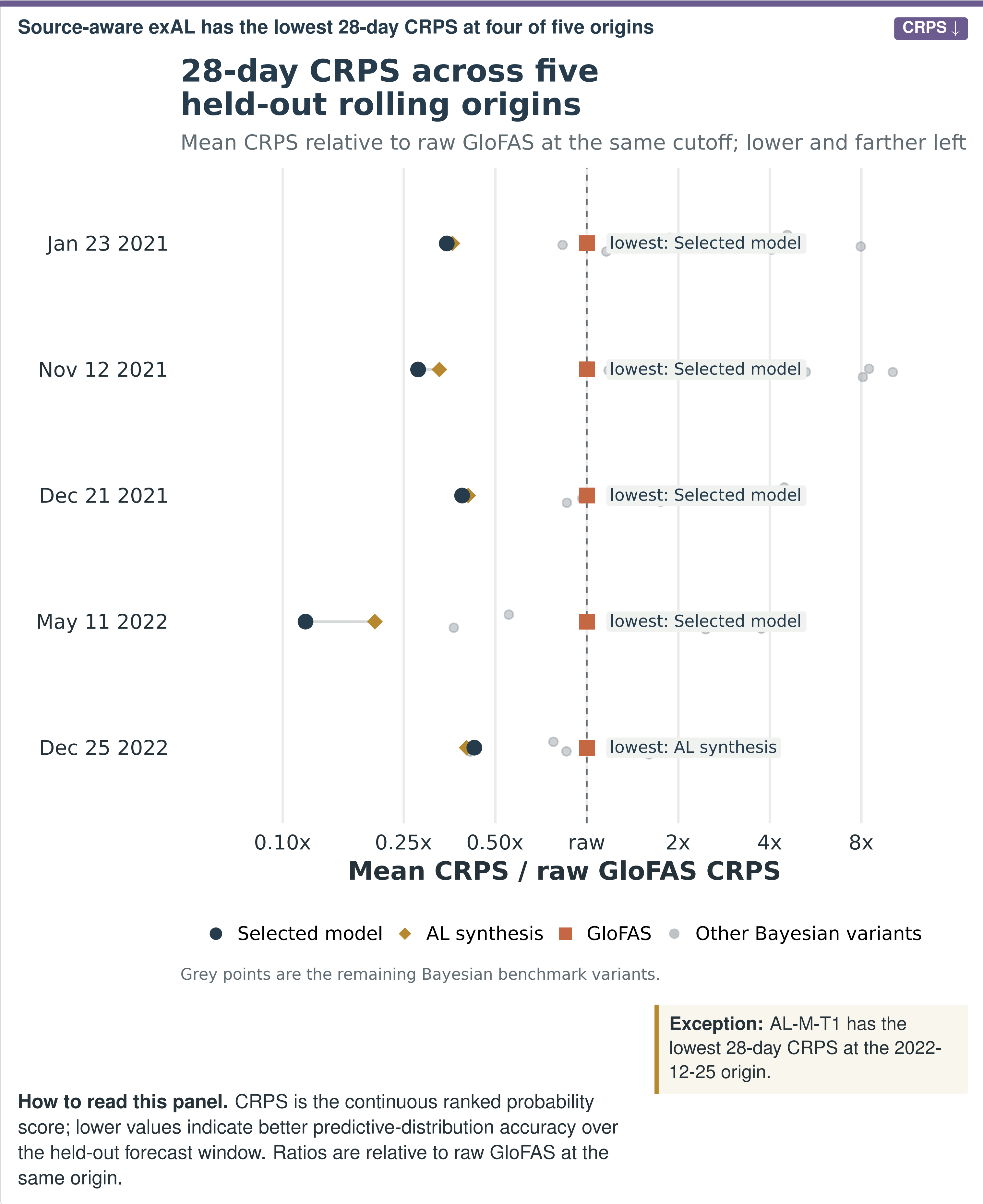
Exogenous adjustment: $\zeta_t \mid \zeta_{t-1}, \psi_{t-1} \sim \mathcal{N}(\lambda \zeta_{t-1} + \mathbf{x}_t^{\psi} \psi_{t-1}, w_t^{\zeta}), \quad 1 \leq t \leq T + K_{\max},$

Covariate-adjustment states: $\psi_t \mid \psi_{t-1} \sim \mathcal{N}(\psi_{t-1}, \mathbf{W}_t^{\psi}), \quad 1 \leq t \leq T + K_{\max}.$

exAL_{p_0} is the extended asymmetric Laplace working likelihood; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is θ_t ; source corrections are δ_t^j ; and (ζ_t, ψ_t) is the exogenous-adjustment block driven by covariates $\mathbf{x}_t$. Component discounting controls the evolution variances.

MAIN EVIDENCE

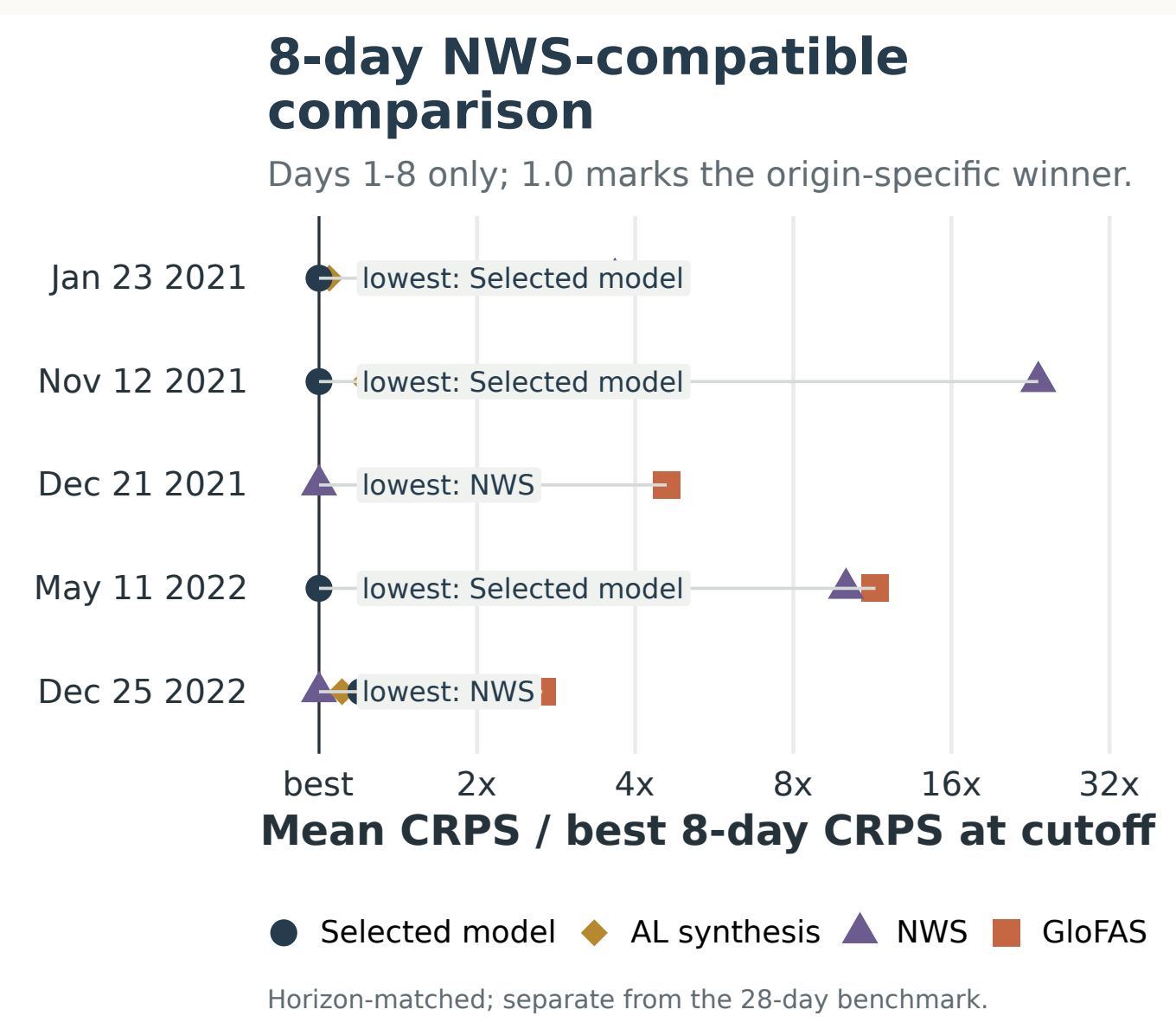
28-day CRPS comparison



Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin; raw NWS is evaluated only in the separate 1–8 day panel.

HORIZON

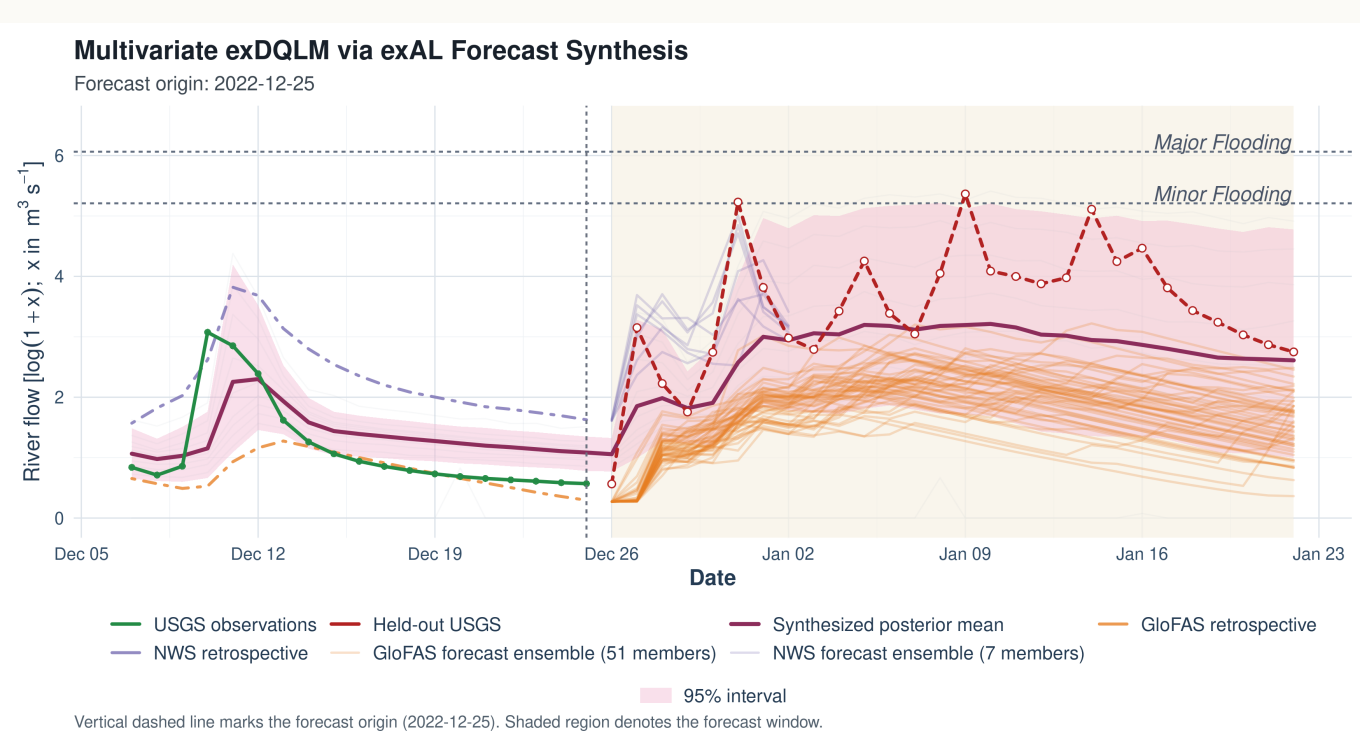
Horizon-matched NWS check



Horizon-matched evidence. Over common days 1–8, exAL-M-T1 has the lowest mean CRPS at three origins and raw NWS at two. The axis is normalized by the origin-specific best 8-day CRPS, so 1.0 marks the winner for that origin; this is separate from the 28-day benchmark.

EXAMPLE

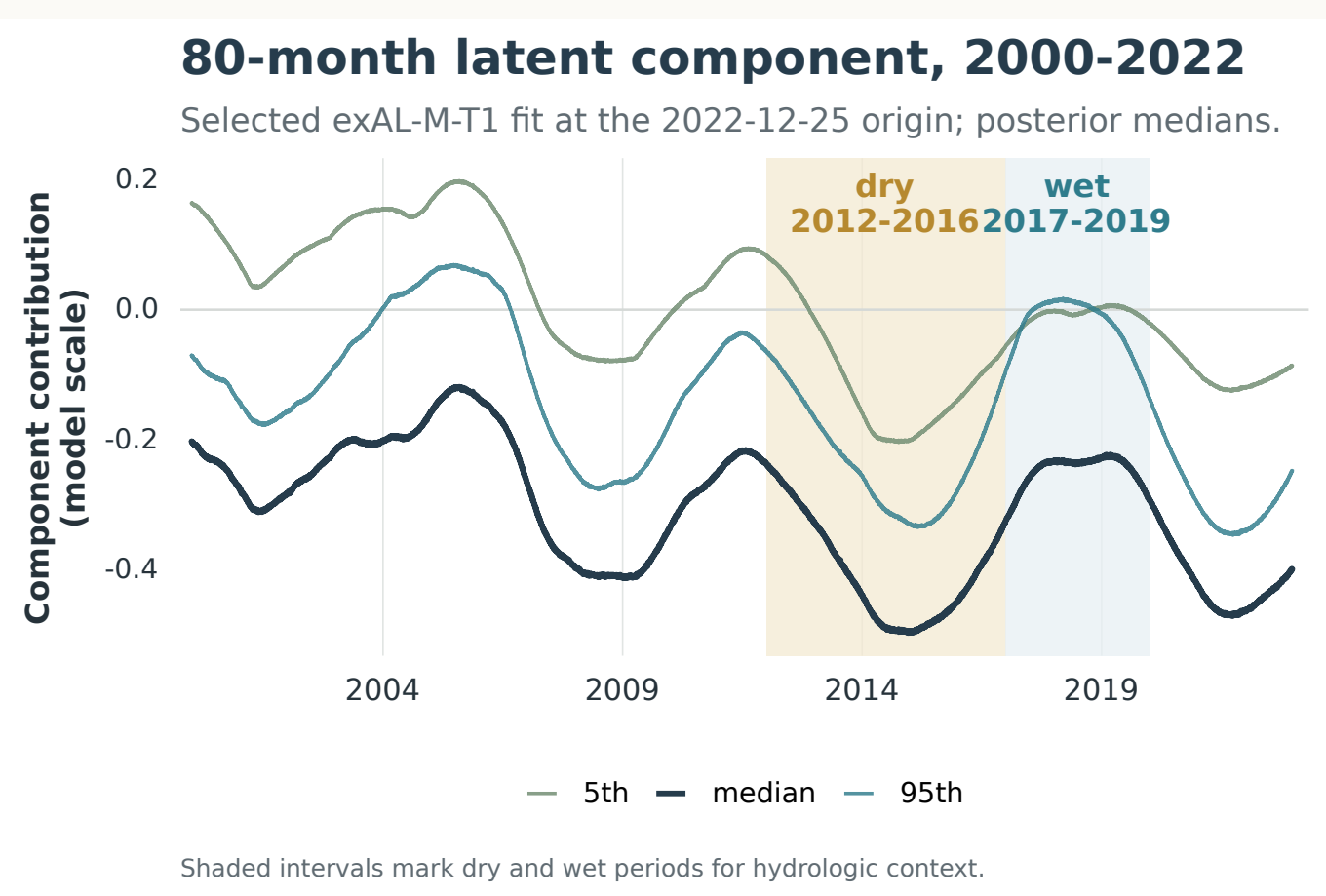
Representative forecast window



Illustrative, not aggregate validation. The selected exAL-M-T1 forecast at the 2022-12-25 origin shows how forecast inputs and posterior uncertainty evolve. At this origin AL-M-T1 has the lowest 28-day CRPS, so this panel is illustrative; aggregate evidence comes from all five origins.

DYNAMICS

80-month latent component



Retrospective diagnostic. From the selected exAL-M-T1 fit at the 2022-12-25 origin. Curves show the 5th, 50th, and 95th target-quantile components over 2000–2022; shading marks dry and wet periods for hydrologic context.

INTERPRETATION

What the evidence supports

- The source-aware exAL model (exAL-M-T1) has the lowest mean 28-day CRPS at four origins; AL-M-T1 is lowest at 2022-12-25.
- Over common days 1–8, exAL-M-T1 is lowest at three origins and raw NWS at two.
- Fitted latent states provide retrospective component diagnostics in addition to forecast-window predictive quantiles.
- The five-window design is a targeted stress test, not a continuous hindcast.

Take-home. Retrospective archives identify source-specific corrections before forecast-product blending. In this five-origin study, the source-aware exAL model leads the 28-day CRPS comparison at four origins and yields interpretable retrospective diagnostics.

